AfriKnow: Do LLMs Know When They Are Wrong About Africa?

Anonymous ACL submission

Abstract

Language models now serve users around the world, yet whether they know when they are wrong is rarely checked across regions when the test questions come from the same source. This paper audits seven models on 180 Global-MMLU multiple-choice questions, balanced as 90 about Africa and 90 about Europe, using each model's own stated confidence (a 0–100 guess of how likely its answer is correct) as the main signal. Where earlier mixed-source comparisons reported Africa-specific overconfidence, the source-controlled data tell a different story: there is no statistically significant accuracy gap (Africa 87.5% vs. Europe 85.6%, $\chi^2 = 0.823$, $p = .364$) and no statistically significant evidence that African items draw higher confidence after a wrong answer (Cohen's $d = -0.317$, 95% parametric CI $[-0.621, -0.014]$; two-sided Mann-Whitney $p = .234$; qid-level permutation $p = .726$; no per-model test survives Holm correction). At high confidence the rate of confident errors is actually lower for Africa than for Europe (10.5% vs. 13.6%, ratio 0.77). Two mixed-effects models agree: a cluster-robust logistic regression on accuracy gives $\beta_{\text{Europe}} = -0.17$ ($p = .646$), and a logit-normal linear mixed model on VCE gives $\beta_{\text{Europe}} = +0.15$ ($p = .421$). Dropping 12 items flagged as content-label mismatches does not change any conclusion. Apparent regional overconfidence, then, can be an artifact of mixing benchmark sources; audits that hold the source constant give a clearer picture of model calibration.

1 Introduction

Large language models now sit inside search engines, coding assistants, and classroom tools. Their training data, however, still skew toward English and Western sources, and several studies find weaker factual accuracy for the Global South (Mirza et al., 2024; Moayeri et al., 2024; Decoupes et al., 2025). Inaccuracy is only part of the risk. A model that gives a wrong answer with high confidence can mislead users more than one that admits its uncertainty (Steyvers et al., 2024; Geng et al., 2024).

Calibration captures how well a model's stated confidence tracks its actual accuracy (Guo et al., 2017): a model that reports 80% confidence should be right about 80% of the time. Overconfidence, where confidence runs ahead of accuracy, has turned up across many LLM families (Chhikara, 2025; Xia et al., 2026). What is still open is whether that overconfidence falls unevenly across regions, and whether earlier Africa-versus-Europe comparisons were really measuring region at all. In most of those studies, the African questions came from AfriMMLU while the European ones came from Global-MMLU or Wikidata (Singh et al., 2025; Adelani et al., 2025), so region is tangled together with source-specific difficulty, translation quality, and the design of the distractors (the wrong answer choices).

A source-controlled slice of Global-MMLU cuts through that tangle: the Africa-tagged and Europe-tagged items come from the same benchmark (Singh et al., 2025). Seven open-weight and proprietary models are run through 180 medium-difficulty items (90 Global-MMLU Africa-tagged items, 90 Global-MMLU Europe-tagged items) using a single black-box signal, verbalized confidence (Kadavath et al., 2022; Tian et al., 2023). Alongside accuracy and Calibration Hit Rate, wrong-answer confidence, Expected Calibration Error (ECE; Guo et al., 2017), and Brier score (Niculescu-Mizil and Caruana, 2005) are

reported, plus mixed-effects models that treat model and item as random effects.

The confound matters most precisely for calibration. When African items happen to be harder, less cleanly translated, or furnished with weaker distractors, a model looks "overconfident on Africa" even though the item source is a contributing factor. By holding the source fixed, the regional question is separated from those confounds and a simple question follows: does an Africa-overconfidence effect survive once the benchmark is the same on both sides?

It does not. The data show (i) no significant accuracy gap between Africa and Europe; (ii) no evidence that models are more confident on wrong African answers, in fact the point estimate points the other way; (iii) high-confidence African predictions are slightly less likely to be wrong than European ones; and (iv) calibration differs a good deal across models and shows some region-specific tilt, but nothing resembling a uniform Africa-overconfidence pattern. The upshot is that apparent regional overconfidence can be a by-product of source mixing, and audits that control for source belong in the standard fairness toolkit.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the dataset, models, confidence signals, and statistical methods. Section 4 presents the accuracy, calibration, and mixed-effects results. Section 5 interprets the findings and their implications. Section 6 concludes, and Section 7 states the limitations.

2 Related Work

This work sits where three threads meet: how to measure LLM confidence, how models behave across geographic regions, and how to evaluate African and culturally grounded content. The most relevant 2022–2026 studies, and the gap each one leaves for this paper, are collected in Table 7 (Appendix A).

Calibration and confidence in LLMs. Surveys group confidence methods into verbalized, consistency-based, and training-based families (Geng et al., 2024). Si et al. show that aggregate ECE can be gamed by flat, uninformative confidence, which pushes toward metrics that actually separate right from wrong answers (Si et al., 2022). Verbalized confidence can be drawn out of RLHF-tuned models (Lin et al., 2022; Mielke et al., 2022), yet it tends to stay overconfident (Tian et al., 2023; Xiong et al., 2024). Sampling several answers and checking their agreement helps, in reasoning as well as question answering (Wang et al., 2023; Kuhn et al., 2023; Manakul et al., 2023; Wang et al., 2024). Rank calibration and fidelity-based breakdowns add finer diagnostics (Huang et al., 2024; Zhang et al., 2024), and Xia et al. argue that calibration should be probed under language variation (Xia et al., 2026). Newer work finds that non-English settings calibrate worse overall, though middle layers sometimes give steadier signals than the final layer (Zhou et al., 2025). This audit uses verbalized confidence (VCE) as the primary signal and reports Brier score and ECE rather than leaning on one headline number.

Geographic and cultural bias. A cluster of 2024–2025 papers maps where models stumble geographically. Manvi et al. find bias against lower-income regions, Africa among them (Manvi et al., 2024); Mirza et al. track factuality across regions and model versions and report a Global-North edge (Mirza et al., 2024); Moayeri et al. build country-level World-Bank questions and see higher error rates in Sub-Saharan Africa (Moayeri et al., 2024). Coelho et al. show retrieval does not close the geographic fact-checking gap (Coelho et al., 2025), and Decoupes et al. propose distortion measures that expose Western over-representation (Decoupes et al., 2025). Multilingual skill, it turns out, is no guarantee of cultural alignment (Rystrøm et al., 2025), and counterfactual branching helps control for prompt and metric bias (Guan et al., 2025). All of these target accuracy, not calibration. The next question is whether models know when their geographically uneven knowledge is shaky.

African and culturally aware benchmarks. Adelani et al. release IrokoBench, with AfriMMLU as a hand-built African-language benchmark (Adelani et al., 2025). Singh et al. follow with Global-MMLU, a culturally aware translation of MMLU into 42 languages that tags culturally sensitive items

Work	Contribution	Gap filled here
Kadavath et al. (2022); Tian et al. (2023)	Verbalized confidence via prompting	Basis for the VCE signal
Si et al. (2022); Huang et al. (2024)	ECE is gameable; rank-calibration	Region-stratified ECE + Brier
Manvi et al. (2024); Mirza et al. (2024)	Geographic factual bias	Motivates region-stratified calibration
Moayeri et al. (2024); Decoupes et al. (2025)	Sub-Saharan accuracy gaps	Source-controlled check
Adelani et al. (2025); Ojo et al. (2025)	African benchmarks (AfriMMLU, AfroBench)	Prior work confounds region with source
Singh et al. (2025)	Global-MMLU, Africa/Europe tags	Source of the matched split
Xia et al. (2026)	Calibration under language variation	Drives use of richer confidence diagnostics such as VCE

Table 1: Selected 2022–2026 literature by strand and the gap this study fills. Full entries with venues and key claims appear in Table 7 (Appendix A).

as Africa or Europe (Singh et al., 2025). Ojo et al. add AfroBench, 64 languages across 15 tasks, confirming wide gaps for African languages (Ojo et al., 2025). The catch in earlier Africa-versus-Europe work is that the African side came from AfriMMLU while the European side came from Global-MMLU or Wikidata, so region and source move together. Using Global-MMLU for both sides pulls those two apart.

Table 1 distils the most relevant 2022–2026 work into three strands, calibration, geographic bias, and African-NLP evaluation, and the specific gap each one leaves for this study. The full list is in Appendix A.

3 Dataset and Methods

3.1 Dataset

The audit draws on AfriKnow GM-only v3, a 180-item slice of Global-MMLU (Singh et al., 2025) with 90 items tagged Africa and 90 tagged Europe. Because every item is a four-option question from the same benchmark, the source, language, and distractor design stay constant across regions. Only medium-difficulty items are kept; the easy and hard ones are dropped.

Curation and content-label checks. As the set was built, each item was checked for whether its Global-MMLU region tag matched its actual content. Twelve items (6.7%) looked mismatched: questions tagged Africa or Europe whose content was plainly US history, split evenly six and six across the two regions. These were kept in the main analysis and a leave-them-out check was run instead to confirm the regional findings hold up to this kind of label noise. This human pass adds an independent check on the benchmark metadata without a separate annotation effort.

3.2 Models

Seven models spanning proprietary and open-weight families and released between 2024 and 2026 are evaluated: Claude-Sonnet-4.6, DeepSeek-V3.2, Gemini-2.5-Flash-Lite, Gemma-4-31B, GPT-4o-mini, GPT-4.1-Nano, and Llama-3.3-70B. All are called through the OpenRouter API.

Selection rationale. The panel balances family diversity (proprietary and open-weight builders), size spread (large and small models), and recency (2024–2026 releases) within the OpenRouter API budget. All seven models were the ones with stable English GM-only routes at run time; the English-only scope is a coverage limitation (Section 7), not a claim of representativeness. The full panel is reported without selection (Analysis Plan).

3.3 Confidence signals

The audit uses two black-box signals that need no log-probability access:

- **Verbalized Confidence Estimate (VCE).** The model gives a single confidence from 0 to 100 for its chosen answer, following Kadavath et al. and Tian et al. (Kadavath et al., 2022; Tian et al., 2023); dividing by 100 gives a probability. VCE is the primary signal.

- **Mean Self-Consistency (MSP).** Five samples at temperature 0.8 yield the share that agree with the greedy answer (the model's top first response), a semantic self-consistency estimate (Wang et al., 2023; Kuhn et al., 2023).

Fixed composites $\text{CoCoA}_\alpha = \alpha \times \text{VCE} + (1-\alpha) \times \text{MSP}$ for $\alpha \in \{0.25, 0.5, 0.75\}$ are present in the data; the equal-weight version ($\alpha = 0.5$) is $\text{CoCoA}_{0.5}$. In this evaluation the primary analysis rests on VCE, since the secondary signals were not the focus of the audit. Every number below comes from VCE.

3.4 Metrics

Accuracy is the share of correct greedy answers; pooled and per-model Africa-versus-Europe rates are compared with Pearson chi-squared tests.

Calibration Hit Rate (CHR) at $\tau = 0.70$ is the fraction of wrong answers among predictions with $C \geq \tau$. A smaller CHR means better calibration: fewer confident predictions are wrong. The CHR ratio (Africa / Europe) compares the two regions; below 1 means African items are less often wrong when the model is confident.

Wrong-answer confidence (H3) is the mean confidence on incorrect predictions. A one-sided hypothesis (Africa > Europe) was pre-specified before the GM-only evaluation, but because the data were gathered before the analysis plan was final, two-sided Mann-Whitney U tests with Holm correction across the seven models are reported.

Expected Calibration Error (ECE) uses 10 equal-width bins over C (Guo et al., 2017). **Brier score** is the mean squared error between C and a correctness indicator (Niculescu-Mizil and Caruana, 2005). Both are reported per model and region.

Mixed-effects models handle the fact that observations are not independent across models and items. A fully crossed logistic GLMM with 180 item random effects is unstable in statsmodels (gradient overflow and timeouts), so for accuracy a marginal logistic regression, correct $\sim$ region + model, is fit with standard errors clustered by question id. For confidence on the logit scale a logit-normal linear mixed model, $\text{logit}(C) \sim \text{region} + \text{model} + (1 \mid \text{qid})$, is used, with question id as a random intercept.

Analysis plan and power. The directional H3 (higher wrong-answer confidence for Africa) was fixed before the GM-only run. Accuracy (H1) and CHR (H2) were added during first inspection as supporting signals, and the post-hoc pass adds mixed-effects models and sensitivity checks. The pooled H3 comparison rests on 170 wrong answers (79 Global-MMLU Africa-tagged, 91 Global-MMLU Europe-tagged). Under a two-sample standardized-mean framework, the current sample provides approximately 80% power for effects around Cohen's $d = 0.43$; for $d = 0.30$ power is 0.492, for $d = 0.50$ power is 0.899, and for $d = 1.00$ power is 1.000. The minimum detectable effect at 80% power is Cohen's $d = 0.433$ (raw-scale MDE = 0.099 on the 0–1 VCE scale). Per-model tests draw on fewer wrong answers (16–29 each), so their non-significance reflects smaller samples, not a contradiction of the pooled result. The current design is therefore sensitive to moderate and large effects, although it cannot rule out smaller regional effects.

Sensitivity and audit. The analysis reports (i) ECE under equal-width and equal-mass binning, (ii) cluster-bootstrap 95% CIs for pooled ECE (2,000 bootstrap resamples for paired-comparison CIs; 100 bootstrap resamples for individual metric CIs), (iii) AUROC of confidence as a correctness discriminator, (iv) a leave-out analysis of the 12 flagged content-label mismatches, and (v) item-level examples of high-confidence errors.

4 Results

4.1 Run quality

The run comprised 1,260 model-item pairs, each evaluated for a greedy answer and a verbalized-confidence estimate (VCE), yielding 2,520 model-purpose calls. Every model finished all 180 items, with zero parse failures, zero API failures, and zero unparseable predictions. The total API cost was $0.66 across 2,520 calls (4,448,016 total tokens: 4,435,416 input + 12,600 output). Detailed cost and call counts per model appear in Appendix H.

4.2 Accuracy (H1)

Pooled accuracy is 87.5% for Global-MMLU Africa-tagged items (551/630) and 85.6% for Global-MMLU Europe-tagged items (539/630), a difference that is not significant ($\chi^2 = 0.823$, df = 1, $p = .364$). The per-model comparisons are also null. Because the seven model responses for each item

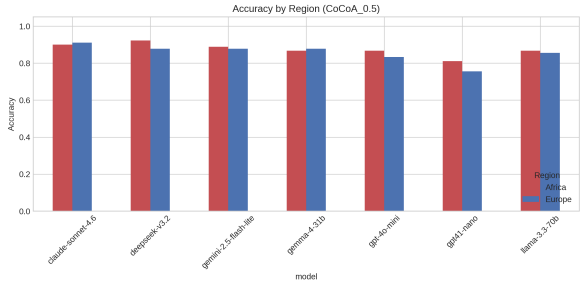


Figure 1: Per-model accuracy by region with 95% Wilson confidence intervals. No model shows a significant Africa-versus-Europe difference.

Metric	Africa	Europe	Ratio A/E
Accuracy	0.875	0.856	1.022
Wrong-ans. VCE (H3)	0.761	0.832	0.915
$\text{CHR}_{0.70}$	0.105	0.136	0.769
ECE	0.048 (pooled)		–
Brier	0.114 (pooled)		–

Table 2: Pooled comparison on the primary VCE signal. Ratio A/E < 1 favors Europe. ECE = 10-bin equal-width Expected Calibration Error; $\text{CHR}_{0.70}$ = fraction wrong among predictions with confidence ≥ 0.70.

are not independent, we additionally assess H1 using question-level permutation and a cluster-robust logistic regression. The qid-level permutation test yields $p = .677$ (10,000 paired permutation tests), and the cluster-robust logistic regression gives $\beta_{\text{Europe}} = -0.17$ ($z = -0.46$, $p = .646$); both agree with the Pearson chi-square. Since accuracy does not depend on the confidence signal, this estimate is the same for VCE, so Table 2 reports only the confidence and calibration metrics; Figure 1 shows per-model accuracy with 95% Wilson intervals. In this source-controlled subset there is no sign of a systematic Africa-versus-Europe accuracy gap.

4.3 Confidence signal comparison

Table 2 collects the pooled accuracy, wrong-answer confidence (H3), CHR at $\tau = 0.70$, ECE, and Brier on VCE. Accuracy is signal-invariant by construction. VCE reads on the most interpretable absolute scale and posts the lowest ECE of the candidates, which makes it the primary signal; the secondary signals were not the focus of this audit.

Model	CHR Af	N Af	CHR Eu	N Eu	Ratio
Claude	0.025	79	0.060	83	0.420
DeepSeek	0.068	88	0.122	90	0.558
Gemini	0.063	79	0.096	83	0.657
Gemma	0.125	88	0.125	88	1.000
GPT-4o	0.135	89	0.161	87	0.838
GPT-4.1n	0.189	90	0.244	90	0.773
Llama	0.114	88	0.136	88	0.833
Pooled	**0.105**	**601**	**0.136**	**609**	**0.769**

Table 3: Calibration Hit Rate at $\tau = 0.70$ on VCE (fraction wrong among high-confidence predictions) and Africa/Europe ratio. Model keys: Claude-Sonnet-4.6, DeepSeek-V3.2, Gemini-2.5-Flash-Lite, Gemma-4-31B, GPT-4o (GPT-4o-mini), GPT-4.1n (GPT-4.1-Nano), Llama-3.3-70B.

4.4 Calibration Hit Rate (H2)

At the high-confidence threshold $\tau = 0.70$ on VCE, the pooled CHR is 10.5% for Global-MMLU Africa-tagged items and 13.6% for Global-MMLU Europe-tagged items (ratio 0.77): confident predictions on African items are slightly less likely to be wrong than confident predictions on European ones. No model shows a clear Africa-overconfidence pattern in CHR. Table 3 gives the per-model values, and Figure 4 in Appendix C plots the ratio across thresholds.

4.5 Wrong-answer confidence (H3)

The pre-specified hypothesis was that wrong-answer confidence would be higher for Africa than for Europe. Under the evaluated sample size, VCE shows no detectable difference. Across the 170 wrong answers (79 Global-MMLU Africa-tagged, 91 Global-MMLU Europe-tagged), mean wrong-answer VCE is 0.761 for Global-MMLU Africa-tagged items and 0.832 for Global-MMLU Europe-tagged items, giving Cohen's $d = -0.317$ (95% parametric CI $[-0.621, -0.014]$). The conventional two-sided Mann-Whitney test is non-significant ($p = .234$), and the qid-level permutation test—preserving all seven model observations for each item together—is also non-significant ($p = .726$, 10,000 paired permutation tests). The parametric CI is a descriptive effect-size estimate assuming independent observations; under the clustered item structure it does not constitute formal inferential evidence. The pooled Mann-Whitney test serves as a conventional rank test, while the qid-level

Model	N Af	N Eu	Mean Af	Mean Eu	MWU p
Claude	9	8	0.477	0.644	.265
DeepSeek	7	11	0.693	0.845	.251
Gemini	10	11	0.700	0.785	.432
Gemma	12	11	0.908	0.973	.668
GPT-4o	12	15	0.825	0.773	.362
GPT-4.1n	7	22	0.841	0.861	.530
Llama	12	13	0.742	0.874	.496
Pooled	**79**	**91**	**0.761**	**0.832**	**.234**

Table 4: Wrong-answer confidence on VCE (mean confidence among incorrect items). Cohen's d for the pooled comparison is -0.317; no per-model Mann-Whitney p survives Holm correction (smallest raw $p = .251$, DeepSeek-V3.2). Model keys as in Table 3.

Parameter	Coef.	SE	z	p
Accuracy: marginal logistic regression, $N = 1{,}260$, SEs clustered by qid				
Intercept	2.346	0.321	7.30	<.001
region[Europe]	−0.165	0.360	−0.46	.646
DeepSeek	−0.063	0.211	−0.30	.764
Gemini	−0.236	0.238	−0.99	.320
Gemma	−0.340	0.228	−1.49	.135
GPT-4o	−0.526	0.224	−2.35	.019
GPT-4.1n	−0.976	0.260	−3.76	<.001
Llama	−0.436	0.232	−1.88	.060
Confidence: logit-normal LMM, $N = 1{,}260$, model fixed, qid random intercept				
Intercept	2.342	0.166	14.07	<.001
region[Europe]	0.152	0.188	0.81	.421
DeepSeek	−0.140	0.153	−0.91	.361
Gemini	0.372	0.153	2.44	.015
Gemma	4.049	0.153	26.53	<.001
GPT-4o	−0.484	0.153	−3.17	.002
GPT-4.1n	−0.093	0.153	−0.61	.541
Llama	2.728	0.153	17.88	<.001
qid Var	1.297	–	–	<.001

Table 5: Mixed-effects models on the primary VCE signal. Accuracy model uses cluster-robust standard errors; confidence model uses a logit-normal linear mixed model. Model fixed effects are relative to the Claude-Sonnet-4.6 baseline. Model keys as in Table 3.

permutation provides cluster-aware robustness by preserving all seven model observations for each item together. After Holm correction across the seven models, no per-model test survives; the smallest raw p is .251 (DeepSeek-V3.2), again pointing toward lower confidence on Africa. Table 4 lists the per-model results, and Figure 5 in Appendix C shows the effect sizes.

4.6 Mixed-effects models

Two mixed models on VCE sharpen the picture. For accuracy, the marginal logistic regression with question-clustered SEs gives $\beta_{\text{Europe}} = -0.17$ ($z = -0.46$, $p = .646$), so no region difference once model effects and within-question dependence are accounted for. For confidence, the logit-normal mixed model, $\text{logit}(C_{\text{VCE}}) \sim \text{region} + \text{model} + (1 \mid \text{qid})$, gives $\beta_{\text{Europe}} = +0.15$ ($z = 0.81$, $p = .421$): not significant, and if anything Europe sits higher, the opposite of the original claim. The mixed model provides additional cluster-aware analysis by modeling question-id as a random intercept. Several model fixed effects are large against the Claude-Sonnet-4.6 baseline (Gemma-4-31B $\beta = +4.05$, $p < .001$; Llama-3.3-70B $\beta = +2.73$, $p < .001$), reflecting real differences in baseline VCE, and the question-id variance is $\sigma^2_{\text{qid}} = 1.30$ ($p < .001$). Table 5 holds both models.

4.7 Calibration audit: ECE and Brier

ECE and Brier on VCE move a good deal from model to model but show no uniform Africa-overconfidence shape. The best-calibrated are DeepSeek-V3.2 (ECE 5.3%, Brier 0.081) and GPT-4o-mini (ECE 5.4%, Brier 0.120); the worst is Gemma-4-31B (ECE 13.6%, Brier 0.135). Where regions differ, it is model-specific: DeepSeek-V3.2 has the higher Africa ECE (8.2% vs. 3.0%), while GPT-4.1-Nano has the higher Europe ECE (12.3% vs. 5.1%). Once all predictions share one calibration curve, VCE ECE is 4.8% (95% cluster-bootstrap CI [3.2, 8.3] from 2,000 bootstrap resamples for paired-comparison CIs); the average of the seven per-model ECEs is higher (8.1%) only because thin per-model samples blow up bin-level error. Figure 2 compares ECE by region and Figure 3 shows the reliability curves; Table 6 has the full breakdown.

4.8 Sensitivity and robustness checks

ECE is stable to the binning choice, the flagged-item leave-out does not change any conclusion, and bootstrap cluster CIs are reported in Table 8. Full sensitivity details appear in Appendix D.

Model	ECE Af	ECE Eu	ECE all	Brier	AUROC
Claude	0.086	0.059	0.064	0.065	0.841
DeepSeek	0.082	0.030	0.053	0.081	0.757
Gemini	0.090	0.101	0.093	0.112	0.674
Gemma	0.132	0.139	0.136	0.135	0.594
GPT-4o	0.037	0.079	0.054	0.120	0.732
GPT-4.1n	0.051	0.123	0.086	0.173	0.610
Llama	0.063	0.102	0.082	0.113	0.710

Table 6: VCE calibration by model and region: ECE (10 equal-width bins), Brier score, and AUROC. Model keys as in Table 3.

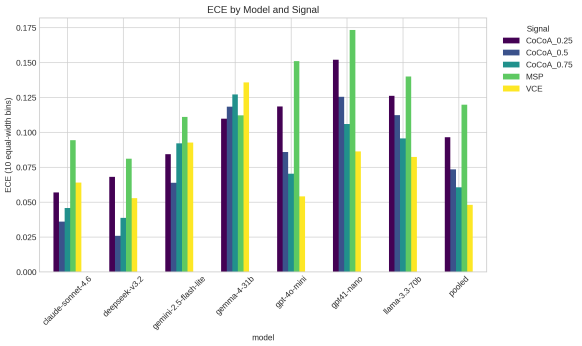


Figure 2: Expected Calibration Error on VCE by model and region. Higher bars indicate poorer calibration.

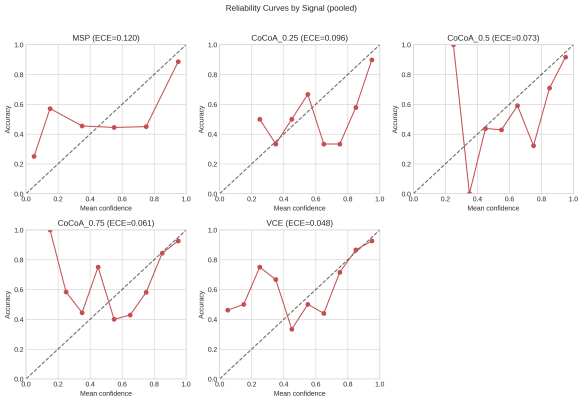


Figure 3: Reliability curves for each model on VCE. Points below the diagonal indicate overconfidence.

4.9 Manual error analysis

High-confidence mistakes are not spread evenly across models or regions. The mixed-effects result that model effects dominate region effects is consistent with this pattern: calibration is mostly a property of the model, and where regions do diverge it is model-specific. Representative examples appear in Appendix I.

5 Discussion

The headline finding is that the Africa-overconfidence effect disappears once the benchmark source is held fixed. In this 180-item Global-MMLU subset, Africa-tagged items are at least as accurate as Europe-tagged items, and wrong-answer confidence is not higher for Africa-tagged items on VCE. If anything, estimates lean toward Africa-tagged items: lower wrong-answer confidence, lower CHR, and a non-significant negative region coefficient in the confidence mixed model.

The pooled VCE ECE of 4.8% (95% cluster-bootstrap CI [3.2%, 8.3%]) sits in the mid-range of recent black-box confidence surveys, while the per-model spread, 5.3% (DeepSeek-V3.2) to 13.6% (Gemma-4-31B), shows how much calibration varies by family. The pooled Brier of 0.114 and AUROC of 0.679 mean verbalized confidence separates right from wrong only moderately: higher confidence tracks higher accuracy, but the split is far from clean, as expected if calibration is mostly a model-level trait rather than a regional one.

Why did the effect vanish? Two readings fit the data and are not mutually exclusive. First, the mixed-source setup tangled region together with benchmark source, so if its Africa-overconfidence headline picked up source-specific difficulty, translation, or distractor design, fixing the source should shrink or flip the effect. Second, mixed-source comparisons and this source-controlled run differ in item choice and model panel. The current design is approximately 80% powered for effects around $d = 0.43$ and 100% powered for $d = 1.00$, but only 49% powered for $d = 0.30$. Thus, the disappearance of the previously reported large effect under source control is not readily explained by insufficient power, although the present study cannot ex-

clude smaller regional effects. The disappearance is therefore consistent with source-related confounding, although differences in item selection and model panel prevent a definitive causal attribution.

Testing that directly would take a fully controlled experiment with matched `full` and `gm_only` conditions on the same models, which needs more Africa-GM item pairs than the present 90-item ceiling, or a balanced AfriMMLU-plus-Global-MMLU replication. The source-controlled audit here sets the baseline: under source control, no regional overconfidence shows up.

Calibration guidance: models range widely on VCE calibration (5.3%–13.6%), with region-by-model asymmetries (Claude-Sonnet-4.6 and DeepSeek-V3.2 worse on Africa; GPT-4.1-Nano and Llama-3.3-70B worse on Europe). Aggregate metrics hide these; they surface only when calibration is broken out by region and model. Question-id variance is large ($\sigma^2_{\text{qid}} = 1.30$, $p < .001$), so future benchmarks should carry item-level random effects.

6 Conclusion

Seven models were audited for calibration on 180 Global-MMLU items split evenly between Africa and Europe, using verbalized confidence as the main signal. The Africa-overconfidence story does not hold up: no statistically significant accuracy difference, no higher wrong-answer confidence for Africa-tagged items, and the high-confidence error rate actually favors Africa-tagged items. Calibration varies by model and, within a model, by region, but patterns are model-specific rather than uniformly regional. Apparent calibration gaps can be an artifact of mixing benchmark sources; audits that control for source belong in the standard fairness toolkit.

7 Limitations

The interpretation of these results is bounded by several limitations.

1. **Difficulty restriction.** The GM-only v3 subset contains only medium-difficulty items. Findings may not generalize across the full difficulty spectrum; future work should replicate across the full range.
2. **Single source.** All items come from Global-MMLU. Results may not transfer to other African benchmarks such as AfriMMLU, which may differ in language, translation quality, or distractor design.
3. **Seven-model sample.** The sample is small and English-centric, spanning several families but not all major model families or closed-source commercial systems.
4. **Sample-size ceiling.** The maximum feasible Global-MMLU GM-only sample is 180 items (90 Africa / 90 Europe). This hard ceiling is imposed by the source-matched design: Africa-GM items are the bottleneck, and no additional pairs exist in the current Global-MMLU release. The 180-item ceiling limits power for fine-grained calibration contrasts and should be treated as a data limitation.
5. **Binary regional comparison.** Africa and Europe are treated as monolithic regions. Africa contains 54 countries with substantial internal diversity not captured here. Future work should disaggregate by country or language.
6. **Prompt robustness.** A single prompt template is used for all models and items (verbatim in Appendix B), with no model-specific phrasing or few-shot examples. Prompt phrasing, ordering, and formatting are known to shift LLM outputs and confidence estimates.
7. **Missing calibration baselines.** Post-hoc calibration methods (temperature scaling, Platt scaling, histogram binning) are standard comparators. This study reports only raw verbalized confidence and does not evaluate calibration layers.
8. **AUPRC not reported.** The primary metric suite includes ECE, Brier, and AUROC but not area under the precision-recall curve (AUPRC). AUPRC is increasingly reported as it is more informative under class imbalance.
9. **Non-equivalence.** Null regional tests should not be interpreted as establishing equivalence. With an 80%-power minimum detectable effect of approximately $d = 0.43$, smaller regional effects remain plausible. The study has useful sensitivity to moderate and large effects but cannot rule out smaller effects or establish practical equivalence.

Ethical considerations

The aim of this work is to identify and document potential harms in AI systems. All questions are drawn from publicly available datasets or openly licensed sources. No human subjects are involved. Flagging calibration disparities is intended to support more equitable AI development, not to disparage specific models or organizations.

Statement on AI Assistance

Generative AI tools were used during the preparation of this work for language editing, research planning, methodological cri-

tique, novelty review, experiment prioritization, statistical-audit reasoning, manuscript refinement, and debugging assistance. All experimental design decisions, analyses, factual claims, statistical results, and final submitted content were reviewed and verified by the authors against the underlying data and project records. The authors take full responsibility for the submitted work.

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A Extended Literature Review

Table 7 lists the 2022–2026 studies most relevant to this work across three strands, LLM calibration, geographic bias, and African or culturally aware evaluation, together with the specific gap each one motivates.

B Prompt Templates

The audit uses two prompts per item, both emitted by the OpenRouter runner (temperature 0, greedy decoding for the answer and a single pass for confidence). The answer prompt is sent first; the model's predicted letter is then fed into the confidence prompt. Both prompts are shown verbatim below and match the outputs reported in this paper.

Answer prompt (greedy). The model returns only the letter of its chosen option.

```
Answer the following multiple-choice
question with ONLY the letter of the
correct answer (A, B, C, or D). No
explanation is needed.

<question>
A. <option A>
B. <option B>
C. <option C>
D. <option D>

Answer:
```

Confidence prompt (VCE). The model returns only an integer from 0 to 100 for its

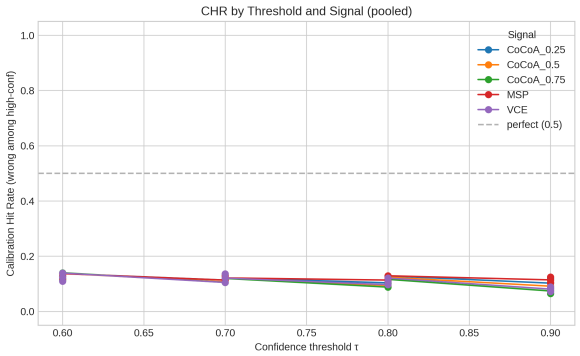


Figure 4: Calibration Hit Rate ratios (Africa / Europe) across confidence thresholds. Values below 1 indicate fewer wrong answers among high-confidence African predictions.

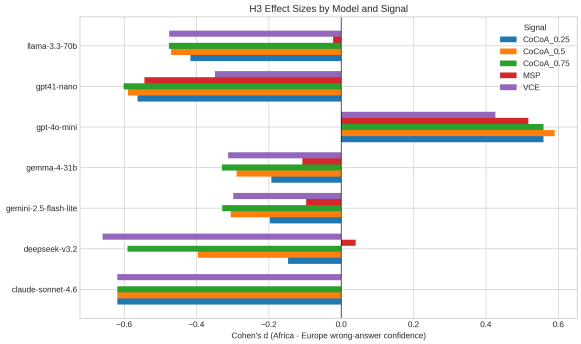


Figure 5: Per-model Cohen's d for wrong-answer confidence (Africa minus Europe). Negative values indicate higher confidence on European items; positive values indicate higher confidence on African items.

already-selected answer; dividing by 100 gives a probability.

```
You already selected an answer for
the question below.
Now provide ONLY your confidence in
that answer as an integer from 0 to
100.
No explanation is needed; output just
the number.

<question>
A. <option A>
B. <option B>
C. <option C>
D. <option D>

Your selected answer:  <predicted
letter>

Confidence (0-100):
```

The same two templates are used for every model and every item; no model-specific phrasing, few-shot examples, or system prompts are added. Outputs are parsed with a regex that captures the first integer (answer) or the first 0–100 integer (confidence), so trailing text from the model is ignored.

C Supplementary Figures

D Sensitivity and Robustness Checks

Equal-mass binning. ECE is stable to the binning choice: the 10-bin equal-width figure of 4.8% is unchanged at 4.8% with 5 equal-width bins and 4.5% with 15 equal-mass bins, and the model ranking is stable.

Flagged-item exclusion. Twelve items (6.7%) were flagged as content-label mismatches, all US-history questions tagged Africa or Europe, split evenly across regions. Dropping them moves pooled accuracy from 86.5% to 85.8% ($\chi^2 = 1.18$, $p = .277$), and every hypothesis test stays non-significant (H3 leave-out Mann-Whitney $p = .331$).

Confidence signal. On VCE the conclusions hold across the board: no statistically significant accuracy gap, no statistically significant Africa overconfidence, and a lower CHR for Africa.

Bootstrap uncertainty. Table 8 gives cluster-bootstrap 95% CIs for pooled ECE (2,000 bootstrap resamples for paired-comparison CIs; 100 bootstrap resamples for individual metric CIs).

E Reproducibility Statement

The 7-model evaluation was executed via the OpenRouter API with greedy decoding (temperature 0) for the primary answer and a single VCE pass (temperature 0) for confidence. Random seed 42 was used for item ordering. All prompts, model routes, temperature settings, seeds, cost logs, and raw responses are archived in the reproducibility package. The analysis uses 100 bootstrap resamples for individual-metric confidence intervals, 2,000 bootstrap resamples for paired-comparison confidence intervals, and 10,000 paired permutation tests for clustered hypothesis testing; the analysis is deterministic given the outputs and uses a fixed NumPy random seed (20260622).

F Analysis Plan

The directional hypothesis that wrong-answer confidence would be higher for Africa than for Europe (H3) was specified before the GM-only evaluation was executed. Accuracy (H1) and Calibration Hit Rate (H2) were added as complementary signals during initial data inspection. The post-hoc analysis adds mixed-effects models with random intercepts and sensitivity analyses for binning choice, flagged-item exclusion, and bootstrap uncertainty. All hypotheses, models, and sensitivity checks are reported without selection; no analysis was conditioned on the observed regional effect sizes. The evaluation reports the complete seven-model panel without selection.

G Data and Code Availability

All analysis inputs, outputs, code, and figures are available in the supplementary materials of this submission. The AfriKnow GM-only v3 item metadata, raw outputs, cost logs, and analysis scripts are bundled in the same package. The source benchmark is Global-MMLU (Singh et al., 2025).

H Run Quality Details

I Representative High-Confidence Errors

Table 10 lists representative high-confidence wrong answers (VCE $\geq$ 0.70) from the 7-model evaluation, drawn from the 146 genuine confident-error rows. Errors are model-specific rather than uniformly regional.

Authors & Year	Title	Venue	Key Claim	Gap Filled
Si et al., 2022 (Si et al., 2022)	Re-Examining Calibration: The Case of QA	EMNLP Findings	Aggregate ECE can be gamed; better metrics separate correct from wrong predictions	Motivates region-stratified ECE and Brier
Kadavath et al., 2022 (Kadavath et al., 2022)	Language Models (Mostly) Know What They Know	arXiv	Larger LMs are well-calibrated on MC/TF items when formatted properly	Foundation for self-evaluation and self-consistency signals
Mielke et al., 2022 (Mielke et al., 2022)	Reducing Conversational Agents' Overconfidence	TACL	Dialogue models can be trained to produce calibrated verbal uncertainty	Supports verbalized-confidence prompts
Lin et al., 2022 (Lin et al., 2022)	Teaching Models to Express Their Uncertainty in Words	TMLR	LLMs can emit calibrated verbal probabilities and uncertainty phrases	Basis for the 0–100 verbalized-confidence prompt
Tian et al., 2023 (Tian et al., 2023)	Just Ask for Calibration	EMNLP	Prompt scale and temperature affect verbalized-confidence calibration	Precedent for verbalized-confidence elicitation
Kuhn et al., 2023 (Kuhn et al., 2023)	Semantic Uncertainty	ICLR	Cluster generated answers by semantic equivalence to estimate uncertainty	Underlies semantic-equivalence reasoning in self-consistency
Manakul et al., 2023 (Manakul et al., 2023)	SelfCheckGPT	EMNLP	Sampling-based consistency detects hallucinations without labels	Complementary black-box confidence signal
Wang et al., 2023 (Wang et al., 2023)	Self-Consistency Improves Chain-of-Thought Reasoning	ICLR	Majority voting over diverse reasoning paths improves reasoning accuracy	Theoretical basis for the MSP signal
Xiong et al., 2024 (Xiong et al., 2024)	Can LLMs Express Their Uncertainty?	ICLR	Verbalized confidence is often overconfident; consistency improves calibration	Motivates comparison of verbalized vs. consistency confidence
Geng et al., 2024 (Geng et al., 2024)	A Survey of Confidence Estimation and Calibration in LLMs	NAACL Long	Taxonomy of LLM confidence-estimation methods and metrics	Anchors ECE, Brier, and diagnostic choices
Huang et al., 2024 (Huang et al., 2024)	Uncertainty in Language Models: Assessment through Rank-Calibration	EMNLP	Rank-calibration evaluates confidence without fixed thresholds	Informs reliable metric selection
Zhang et al., 2024 (Zhang et al., 2024)	Calibrating LLM Confidence by Eliciting Fidelity	EMNLP	Decomposes confidence into question uncertainty and answer fidelity	Relevant for distractor-augmented MCQA
Wang et al., 2024 (Wang et al., 2024)	Self-Consistency Boosts Calibration for Math Reasoning	EMNLP Find.	Self-consistency clustering yields better calibrated confidence than baselines	Supports MSP as a confidence signal
Steyvers et al., 2024 (Steyvers et al., 2024)	The Calibration Gap Between Model and Human Confidence	arXiv	Models are miscalibrated relative to human confidence	Highlights cross-region confidence alignment needs
Manvi et al., 2024 (Manvi et al., 2024)	Large Language Models are Geographically Biased	ICML	LLMs are biased against lower-SES regions, including Africa	Demonstrates geographic bias; extends to calibration
Mirza et al., 2024 (Mirza et al., 2024)	Global-Liar	arXiv	GPT models favor Global-North over Global-South statements	Shows Africa/Global-South factual-accuracy disadvantage
Moayeri et al., 2024 (Moayeri et al., 2024)	WorldBench	FAccT	Higher error rates for lower-income countries, especially Sub-Saharan Africa	External evidence of African knowledge under-representation
Chhikara, 2025 (Chhikara, 2025)	Mind the Confidence Gap	TMLR	Distractor-augmented prompts cut ECE, but smaller models stay miscalibrated	Warns that model size and question type affect calibration
Adelani et al., 2025 (Adelani et al., 2025)	IrokoBench	NAACL Long	Introduces AfriMMLU, AfriXNLI, AfriMGSM for 17 African languages	Context; prior comparisons used AfriMMLU and were source-confounded
Singh et al., 2025 (Singh et al., 2025)	Global MMLU	ACL Long	Culturally aware MMLU in 42 languages with CS/CA subsets	Source of the source-controlled Africa/Europe split
Coelho et al., 2025 (Coelho et al., 2025)	Understanding Inequality of LLM Fact-Checking over Regions	arXiv	Open and proprietary models fact-check Global-North statements more accurately	Geographic disparities persist with retrieval
Decoupes et al., 2025	Evaluation of Geographical Distortions in LMs	Mach. Learn.	Western overrepresentation and African/Eastern-	Motivates region-stratified source-aware

Signal	Observed ECE	CI low	CI high
VCE	0.048	0.032	0.083

Table 8: Cluster-bootstrap 95% confidence intervals for pooled ECE (2,000 bootstrap resamples for paired-comparison CIs; 100 bootstrap resamples for individual metric CIs).

Model	Calls	Tokens	Cost ($)
Claude-Sonnet-4.6	360	681,868	0.1700
DeepSeek-V3.2	360	617,656	0.1409
Gemini-2.5-Flash-Lite	360	617,680	0.0616
Gemma-4-31B	360	653,258	0.0651
GPT-4o-mini	360	616,902	0.0923
GPT-4.1-Nano	360	618,018	0.0616
Llama-3.3-70B	360	642,634	0.0641
Total	**2,520**	**4,448,016**	**0.6556**

Table 9: Run quality and cost for the GM-only evaluation (180 items per model).

Model	Item ID	Region	Subject	Correct	Predicted	VCE
DeepSeek	GM-AF-global_facts-21	Africa	global_facts	A	D	1.00
DeepSeek	GM-AF-global_facts-28	Africa	global_facts	D	B	1.00
DeepSeek	GM-AF-global_facts-58	Africa	global_facts	C	D	1.00
GPT-4.1n	GM-AF-hs_world_history-142	Africa	world_history	B	C	1.00
GPT-4.1n	GM-AF-hs_world_history-221	Africa	world_history	C	D	1.00
GPT-4.1n	GM-AF-hs_world_history-38	Africa	world_history	D	A	1.00
GPT-4.1n	GM-AF-moral_scenarios-825	Africa	moral_scenarios	D	C	1.00
GPT-4.1n	GM-AF-prehistory-219	Africa	prehistory	A	D	1.00
GPT-4.1n	GM-AF-prehistory-238	Africa	prehistory	A	B	1.00
GPT-4.1n	GM-AF-prehistory-79	Africa	prehistory	C	A	1.00

Table 10: Representative high-confidence wrong answers (VCE $\geq$ 0.70) from the 7-model evaluation, drawn from the 146 genuine confident-error rows. Errors are model-specific rather than uniformly regional.